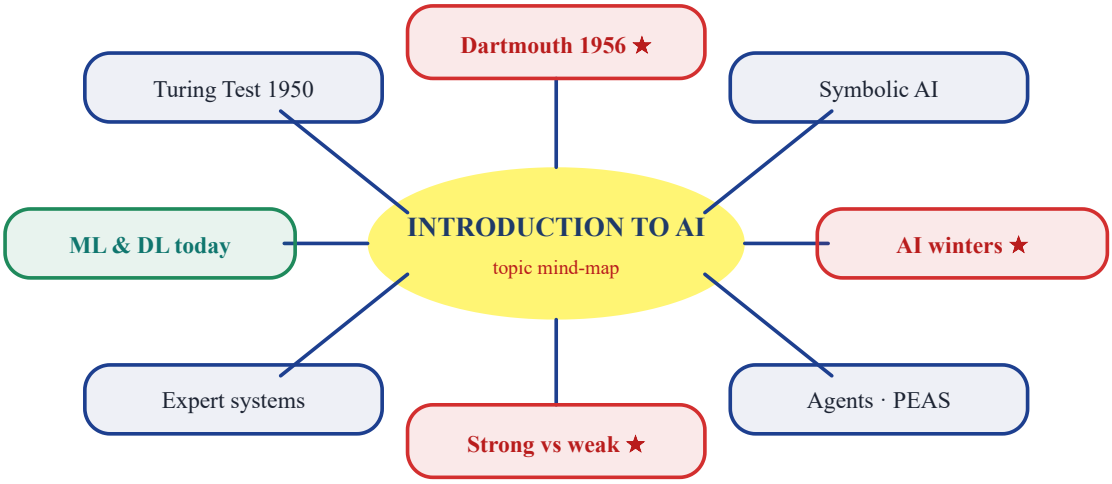
Introduction to AI

Definition · Turing Test · Dartmouth · symbolic vs neural · winters · agents — 12 sheets



↑ red = high-yield, green = the one approach everything now rests on

Syllabus coverage

The AI syllabus lists **introduction to old and new AI techniques** as its first line, before knowledge representation, logic, search, probabilistic reasoning and NLP. This notebook covers that first line: what AI is, how the field arose, and why the modern techniques displaced the old ones.

High-yield list

① AI ⊃ ML ⊃ DL
② Turing Test — Turing, 1950
③ "AI" coined — McCarthy, 1955
④ Field born — Dartmouth, 1956
⑤ Winters: 1974–80, 1987–93
⑥ Four families: symbolic, connectionist, evolutionary, probabilistic
⑦ MYCIN · STRIPS · A* (1968)
⑧ Perceptron 1958 · backprop 1986 · AlexNet 2012

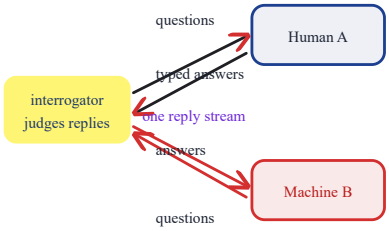
Legend: **yellow** = key term · **red circle** = must-memorise · **purple** = memory trick · **red box** = exam trap · **green** = worked example

Contents

✓ What is AI? definition & Turing Test … p. 2
✓ History & the two winters … p. 3
✓ Old AI: symbolic (1950s–80s) … p. 4
✓ Connectionist AI: neural nets … p. 5
✓ Evolutionary & probabilistic AI … p. 6
✓ Old vs new AI compared … p. 7
✓ Strong AI vs weak AI … p. 8
✓ The agent view, PEAS … p. 9
✓ Current techniques … p. 10
✓ Night-before p. 11 · Practice p. 12

1 What Is AI? Definition & the Turing Test

Definition: **Artificial Intelligence** is the branch of computer science that builds systems able to perform tasks which normally require human intelligence — reasoning, learning, perception, planning. The test is behavioural: **can the system do the task?**, never "does it understand?". The nesting used in almost every paper is **AI ⊃ machine learning ⊃ deep learning**.



text only — no sight, no sound, no prior information

The Turing Test, 1950: Turing set the **imitation game** in *Computing Machinery and Intelligence*: an interrogator holds a written conversation with a hidden human and a hidden machine, one reply stream at a time. The machine **passes if the interrogator cannot reliably tell which one is the machine**. Conditions: **natural language**, **no prior information**, a **third human** as a safety valve.

Exam trap: Passing proves **indistinguishability of behaviour**, never consciousness. **ELIZA** (Weizenbaum, **1966**) fooled many users while understanding nothing; **Chinese room**: Searle, **1980**.

Capability	What it must do	Example	Capability	What it must do	Example
Reasoning	solve new problems	chess engine, A*	Language	read and write text	translator
Knowledge	store facts	semantic network	Perception	read the sensors	speech-to-text
Learning	improve from data	classifier	Planning	pick actions in order	route planner

Memory trick: "Turing tested the **output**, not the brain." Nothing in the test asks the machine to report what it is doing inside — only whether a human can sort the replies into "person" and "machine".

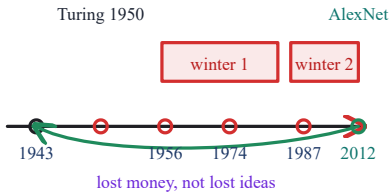
Worked example ✍ is each system AI?: A calculator evaluating 2 + 2 is **not** AI: no inference, no adaptation. A rule-based system returning an infection and a drug from symptoms **is** AI — it reasons over stored knowledge to produce something the user did not supply. A CNN labelling chest X-rays is AI and specifically **machine learning**.

2 How AI Began: History & the Two Winters

Definition: An **AI winter** is a period in which funding, publications and interest collapse because promised results failed to arrive. AI has had **two**: roughly **1974–1980** and **1987–1993**. The pattern repeats — over-promise, under-deliver, funding stops.

Year	Event	Why it matters
1943	McCulloch & Pitts neuron	first mathematical neuron
1950	Turing test, *Mind*	intelligence testable
1951	Minsky & Edmonds, SNARC	first machine tuning weights
1955	McCarthy uses the term "AI"	name precedes field
1956	Dartmouth workshop	birth of the discipline
1958	Rosenblatt's perceptron	first trainable network

Year	Event	Why it matters
1969	Minsky & Papert, *Perceptrons*	one layer's limits
1974	Lighthill report; DARPA cuts	winter 1 begins
1980s	expert-system boom; XCON	commercial peak
1987	Lisp machines unsellable	winter 2 begins
1997	Deep Blue beats Kasparov	search beats a champion
2012	AlexNet wins ImageNet	deep learning mainstream



the field went quiet twice

Winter	Immediate cause	What had gone wrong
1974–80	Lighthill report, UK cuts	translation over-promised; systems did not scale
1987–93	Lisp machines unsellable	expert systems too narrow; Fifth Generation fell short

Exam trap: Three dates get confused: **1950** is the Turing test, **1955** is when McCarthy coined the *term*, and **1956** is the workshop at which the *field* was born.

Memory trick: "Test, name, birth": **1950** test → **1955** name → **1956** birth. A winter starts when a **report** appears — Lighthill 1974, ALPAC 1983.

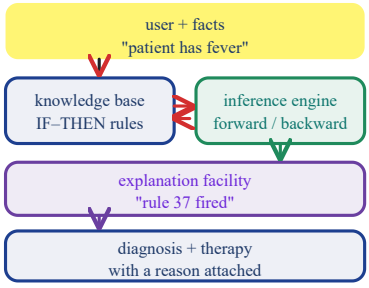
Worked example ✍️ which claim explains which winter?:

1. "The Lighthill report said AI had delivered none of its promises, so funding stopped." → **winter 1**, 1974–1980.
2. "Companies bought Lisp machines whose expert systems were too slow and too narrow." → **winter 2**, 1987–1993.
3. "A single-layer perceptron cannot learn XOR." → not a winter but a **technical limit**, exposed in 1969.

3 Old AI: Symbolic / Rule-Based AI

Definition: **Symbolic AI** — also classical, rule-based or knowledge-based AI — represents intelligence as **explicit symbols and rules** that a program manipulates. Knowledge is written down by people; inference is a formal procedure: deduction, search, or rule chaining.

Family	Core idea	Representative system	Family	Core idea	Representative system
Expert systems	IF–THEN rules + inference engine	MYCIN, DENDRAL, XCON	Knowledge rep.	facts as graphs, frames, rules	semantic network, production rules
Search	explore a state space with a heuristic	BFS, DFS, A* (1968)	Planning	build an action sequence from a goal	STRIPS (1971)
Logic	deduction over predicates	propositional / first-order logic	Belief revision	revise beliefs when facts change	truth-maintenance systems



every expert system has these parts

Exam trap: An expert system **never learns**: every rule was written in advance, so adding one can silently break another — the knowledge-acquisition problem. **Forward chaining** is data-driven (fire rules as facts arrive); **backward chaining** is goal-driven (ask what must be true to prove the goal). STRIPS is **planning**, not logic.

Worked example ✍️ MYCIN, both directions: MYCIN (Stanford, early 1970s) held a few hundred IF–THEN rules on bacterial infections and drug choice. **Forward:** the patient has a fever, so fire every rule whose IF matches — conclusions arrive from the data. **Backward:** the goal is "recommend ampicillin"; ask which conditions must hold, then look for each one. Its advice was rated as acceptable as a specialist's, yet it was never used routinely: **maintaining the rule base cost more than the consultations saved**.

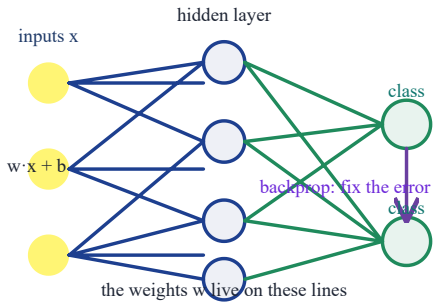
Memory trick: "Symbols are written by **people**, so the bottleneck is human." No rule, no knowledge — and no way to discover a rule the expert never told you.

The one clear commercial success: **XCON** (DEC, 1980) configured VAX systems from a customer order; reports put its savings at tens of millions of dollars a year. Narrow, repetitive, rule-shaped work is what symbolic AI is good at.

4 Connectionist AI: From the Neuron to Deep Learning

Definition: **Connectionist AI** — sub-symbolic, neural or machine-learning AI — builds intelligence out of many simple units wired in layers. Each unit computes a weighted sum of its inputs and passes it through an activation; the knowledge lives not in written rules but in the **weights**, which are fitted to data by an optimisation algorithm.

Year	Milestone	What it added	Year	Milestone	What it added
1943	McCulloch–Pitts neuron	logic gates as a neuron, no learning	1989	LeNet / CNN	one filter shared across an image
1958	Rosenblatt's perceptron	first network trained from data	1997	LSTM	a cell that remembers across sequences
1969	Minsky & Papert	one layer cannot represent XOR	2014	GAN	two networks compete to generate
1986	backpropagation	train many layers, credit layer by layer	2017	Transformer	attention only; basis of language models



every edge carries a weight; training nudges the weights until the answer is right

Exam trap: A single-layer perceptron can learn **AND, OR, NOT** but **not XOR** — XOR is not linearly separable. That 1969 result was never a proof that networks cannot work. And **deep** = many hidden layers, not many neurons in one layer.

Worked example ✍ one threshold unit, then two layers: A unit computes $y = 1$ if $\Sigma w_i x_i \geq \theta$. For **AND** take $w = (1, 1)$, $\theta = 1.5$ → only (1,1) passes; for **OR** take $\theta = 0.5$. For **XOR** a single unit cannot do it, because one straight line cannot separate (0,1),(1,0) from (0,0),(1,1). With two layers it is trivial: hidden units computing OR and AND, then an output unit weighting them (1, 1) with threshold 1 — (0,1) and (1,1) give 1, the rest give 0 ✓.

Memory trick: "Symbolic AI is a **laws book**; connectionist AI is a **trained reflex**." Laws are written and argued with; reflexes are fitted and used without explanation.

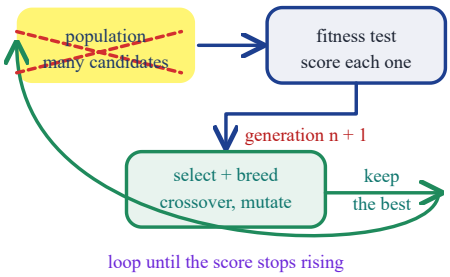
Which architecture for which data: **CNN** for images · **RNN/LSTM** for sequences read in order · **Transformer** when long-range attention matters, which is what language models use · the price of all of them is **data and compute**, paid for in **interpretability**.

5 Evolutionary and Probabilistic AI

Definition: Two more families complete the picture. **Evolutionary AI** keeps a **population of candidate solutions** and breeds the fitter ones — selection, crossover, mutation — exactly as natural selection does. **Probabilistic AI** refuses to be certain: it keeps a **distribution over possibilities** and updates it with Bayes' rule as evidence arrives.

> Bayes only says "most probable" — not true.

Family	Where the knowledge sits	Representative methods	Fails when
Symbolic	in rules written by humans	expert systems, A*, resolution, STRIPS	the rules are missing or wrong
Connectionist	in weights fitted to data	perceptron, CNN, RNN, Transformer	data is thin, or an explanation is required
Evolutionary	in the fittest of a population	genetic algorithm, genetic programming, ant colony	evaluation is slow, or the space is huge
Probabilistic	in conditional probability tables	Bayesian network, HMM, Naive Bayes	the independence assumptions break



a genetic algorithm: the randomness is in the mutations, never in the choice of the fittest

Memory trick: "GA is **guided random**: probability does the mutating, fitness does the choosing." Bayes asks which explanation the **evidence** makes most likely, not which explanation is true.

Exam trap: A genetic algorithm is **not** random guessing: selection is directed and the best candidates are carried unchanged (elitism). In a Bayesian network arrows run from cause to effect and every node carries a conditional probability table; an HMM has hidden states plus emissions and powers speech recognition.

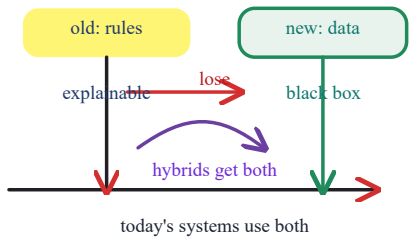
Worked example ✍️ one generation, and one Bayes base rate:

1. **GA:** population {(3, 7), (5, 2), (2, 4)} maximising xy. Fitness: 21, 10, 8. Keep (3,7), copy it twice, mutate the copies slightly → {(3,7), (3,6), (4,8)}. The best product rises from **21 to 32** in one generation — the search is improving, not restarting.
2. **Bayes:** a disease hits 1 % of people; a test finds it in 90 % of the sick and wrongly flags 10 % of the healthy. Of 10 000 people, 100 are sick and 90 test positive; of the 9 900 healthy, 990 test positive. So P(sick | positive) = 90/(90 + 990) = **1/12 ≈ 8.3 %** — the base rate dominates, which is why "the test is 90 % accurate" does not mean a positive result means 90 %.

6 Old vs New AI Techniques, Compared

Definition: "Old AI" is the symbolic, rule-based work of the **1950s–1980s**; "new AI" is the statistical, data-driven work that took over from the **1990s onward**. The dividing line is not "old = wrong, new = right" — it is where the knowledge comes from: **a person wrote it** versus **data supplied it**.

Feature	Old AI	New AI	Feature	Old AI	New AI
Approach	symbolic, rule-based	data-driven, statistical	Data needed	none, if rules complete	large, often labelled
Knowledge from	an expert writes rules	learned from data	Interpretability	**high** — rule shown	low — "black box"
Reasoning	deduction, search	generalisation	Scalability	poor — rules collide	good — data helps
Uncertainty	binary true / false	probabilities	Compute	small machines	GPUs / TPUs
Adaptability	low — means rewriting	high — retrain	Systems	MYCIN, STRIPS, A*	k-means, CNN, LLM



the trade is explainability against scale

Memory trick: "Old AI is **brittle and honest**: it breaks on anything unanticipated but always shows its work. New AI is **flexible and opaque**: it handles anything in distribution but cannot say why."

Common misconception: "Symbolic AI is obsolete" is **false**: SAT and theorem provers, planners, compilers and test-case generators are symbolic systems working today. What changed is the **default** for perception and language, not the existence of reasoning.

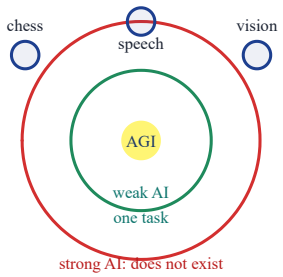
Exam trap: New AI is not "AI plus data" — **knowledge representation** and **search** are still AI whether the symbols were typed by a person or extracted from data. Interpretability is a requirement, not a luxury: in medicine, credit and public administration a system that cannot justify a decision is not deployable at all.

Worked example ✍️ choose the approach: Admit patients whose lab values match **eleven published rules → symbolic expert system**: the rules exist and are auditable. Flag tumours in a new hospital's scans with **no labels yet → deep learning**: there is nothing to write rules from, and images are what CNNs handle. Choose routes for 500 vans from **last year's traffic data → learned prediction**: the answer is a number nobody can write a rule for.

7 Strong AI vs Weak AI

Definition: **Weak AI** (narrow AI, ANI) is built for **one specific task** and displays intelligence inside that task and nowhere else — a chess engine, a spam filter, a face unlock. **Strong AI** (general AI, AGI) would match **humans across every cognitive task**, learn any task from few examples, and form its own goals. Superintelligence (ASI) would exceed the best human performance on everything.

Property	Weak AI (all that exists today)	Strong AI (the goal)
Scope	one task, narrow domain	any task a human can do
Transfer between tasks	nothing carries over	one skill supports the next
Training data	millions of examples for that task	a few examples, or none
Own goals	none — objectives sit in code	forms, revises, pursues its own
Consciousness	not required, not claimed	usually assumed, still undefined
Exists?	**yes — every AI product on sale**	no, never demonstrated



every deployed system sits inside the small circle; nothing is inside the large one

Exam trap: Winning at chess or Go is **not** strong AI: **Deep Blue** (1997) and **AlphaGo** (2016, 4–1 over Lee Sedol) beat world champions at *one* game and were helpless at everything else. The Chinese-room argument (Searle, **1980**) says running the right symbols is not understanding them.

Where the real debate sits: Not on whether machines can beat humans at games — that is settled. It sits on whether understanding can exist **without** consciousness, and on aligning an objective with human intent.

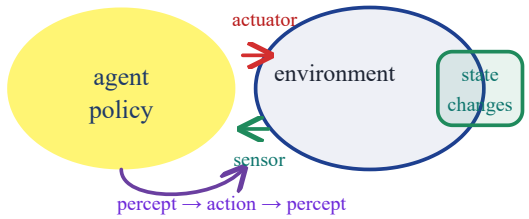
Memory trick: "Weak is **wide but shallow** across data; strong is **deep** across tasks." Today AI has breadth without understanding — a million narrow skills, no mind behind them.

Worked example ✍️ classify four systems: **A chess engine** → weak AI: superb at chess, unable to explain a photograph. **A hospital triage rule base** → weak AI: one narrow task, but it can state which rule fired. **A large language model that writes code and translates** → weak AI: many tasks, yet it needs a new fine-tune for a new domain. **A system that learns one task, carries the skill into a second and sets its own goals** → the **only** strong AI candidate, and none exists.

8 The Agent View: Rational Agents & PEAS

Definition: An **agent** is anything that **perceives its environment through sensors and acts on it through actuators**. An **agent function** maps the whole history of percepts to the next action. A **rational agent** takes the action best for its goals — best **given what it knows**, not best in reality. A goal-oriented system is specified with **PEAS**: Performance, Environment, Actuators, Sensors.

Agent type	How it decides	Weakness	Agent type	How it decides	Weakness
Simple reflex	current percept → fixed rule	loops forever; no memory	Goal-based	searches for a sequence to a goal	only as good as the goal given
Model-based reflex	keeps an internal world model	the model can be stale	Utility-based	maximises a utility score	needs to compare outcomes
Learning agent	improves its own decision rule	can learn the wrong thing	Example: vacuum	suck if dirty, else move right	needs memory to stop looping



the agent–environment loop; almost every AI problem can be drawn this way

Agent	Environment	Actuators	Sensors
vacuum	two dirty rooms	move, suck	position, dirt
drone	airspace, partly observable	rotors, drop	GPS, IMU
spam filter	one mailbox	spam / inbox	headers, body

Exam trap: Rational does **not** mean human-like, and an agent need not have a body — a spam filter is an agent. A simple reflex agent with no memory can never leave a loop, and a utility-based agent differs from a goal-based one by comparing **several goals at once**.

Worked example ✍ the vacuum agent: **Simple reflex:** dirt here → suck, else move right. In a two-room world it walks back and forth *through the dirt it just cleaned* — an infinite loop, having no memory of where it has been. **Model-based reflex:** record which squares are clean — two extra states, no search, loop gone. **Goal-based:** plan the move sequence first. **Utility-based:** add cleaning time and battery to the score.

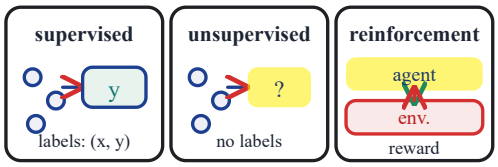
IT link: A monitoring or autoscaling system gets the same four-part description in its design document: what good looks like, where it runs, what it can change, what it can read.

9 Current Techniques: ML, Deep Learning, NLP

Definition: "New AI" is not one technique but a stack. **Machine learning** is the base: systems that fit parameters to data instead of being handed rules. **Deep learning** is machine learning with many layers. **Probabilistic reasoning** handles uncertainty, **evolutionary computation** searches, and **NLP** is the application area that drove the last decade.

Paradigm	Feedback	Standard algorithms	Typical task
Supervised	**labelled** output per input	linear & logistic regression, decision tree, SVM, k-NN	classification, regression
Unsupervised	none — structure discovered	k-means, hierarchical, DBSCAN, PCA, Apriori	clustering, association
Reinforcement	a scalar **reward** after acting	Q-learning, SARSA, policy gradient	games, control, scheduling

Layer	What it contributes
Deep learning	CNN for images, RNN/LSTM for sequences, **Transformer** for language models
Probabilistic	Bayes' rule, Bayesian networks, HMM, Gaussian models
Evolutionary	genetic algorithms, genetic programming, annealing, ant colony
NLP	tokenisation, stemming, lemmatisation, POS tagging, NER, translation



three paradigms, one question: where does the feedback come from?

Memory trick: "A **teacher** gives answers (supervised) · **no teacher**, find groups (unsupervised) · a **critic** gives only a score (reinforcement)."

Exam trap: Supervised learning **requires labelled outputs**; clustering without labels is unsupervised, and reward after an action is reinforcement. **Stemming** chops suffixes mechanically ("studies" → "studi"); **lemmatisation** uses the dictionary ("studies" → "study").

Worked example ✍️ label each scenario:

1. Labelled e-mails marked spam / not spam → **supervised** classification.
2. Grouping shopkeepers by basket similarity, no labels → **unsupervised** clustering.
3. A rover rewarded for distance without falling → **reinforcement** learning.
4. Finding "diaper ⇒ beer" rules from baskets → **unsupervised** association rules (Apriori).
5. Estimating the chance a loan defaults → **supervised**; a 0–1 probability is **logistic** regression.
6. Predicting house prices from size, location, age → **supervised** regression; least squares minimises the **sum of squared residuals**.

Night-Before-the-Exam Sheet

Fact sheet:

AI ⊃ ML ⊃ DL

"AI" coined: McCarthy, **1955**

Winters: **1974–1980**, **1987–1993**

Perceptrons book: **1969**

ELIZA: **1966** · Chinese room: Searle, **1980**

AlexNet: **2012** · Transformer: **2017**

Agent = sensors + actuators

5 agent types: reflex, model-based, goal, utility, learning

Turing Test: Turing, **1950**

Field born: **Dartmouth, 1956**

Perceptron: Rosenblatt, **1958**

backpropagation: **1986**

Deep Blue: **1997** · AlphaGo: **2016**

STRIPS (1971) · A* (1968)

PEAS: Performance, Environment, Actuators, Sensors

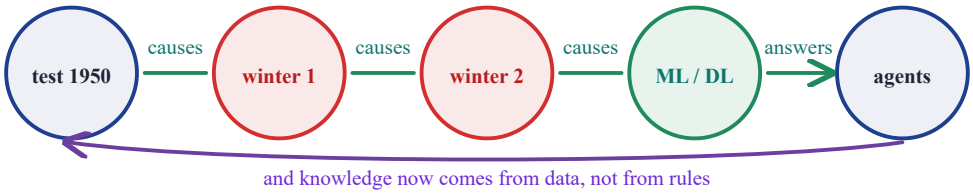
Supervised = labels · RL = reward

Top 12 one-liners for MCQs

1. AI = systems doing tasks that need human intelligence
2. Turing test asks whether replies can be told from a human's
3. McCarthy named AI; Dartmouth 1956 is its birth
4. Two winters, both after over-promising
5. Symbolic AI: rules written by people
6. Connectionist AI: weights fitted to data
7. A single-layer perceptron cannot learn XOR
8. Four families: symbolic, connectionist, evolutionary, probabilistic
9. Weak AI = one task; strong AI = any task, does not exist
10. An expert system never learns and can always explain
11. Forward chaining is data-driven, backward is goal-driven
12. Rational = best action for the goal, given what it knows

Top traps: The Turing test proves behaviour, never understanding · 1950 is the test, 1955 the name, 1956 the birth · expert systems do not learn · symbolic AI is not extinct — provers, planners and compilers still run on it · winning chess or Go is weak AI · supervised needs labels, RL needs reward · stemming is mechanical, lemmatisation is lexical.

All tricks together: "Test, name, birth" (1950 / 1955 / 1956) · "a winter starts with a report" · symbolic AI is a laws book, connectionist AI a trained reflex · old AI is brittle but honest, new AI flexible but opaque · GA is guided random: probability mutates, fitness chooses · a teacher, no teacher, a critic · P-E-A-S.



one test, two winters, then the data-driven turn that produced today's systems

✓ Practice MCQs (exam pattern, 10 marks)

1. The term "AI" was coined in connection with:
(a) the 1943 neuron (b) Dartmouth, 1956
(c) the ARPANET launch (d) Deep Blue, 1997

2. The Turing Test checks whether:
(a) all maths is solved (b) the program halts
(c) a human cannot tell it apart (d) a race is won

3. Breeding a population by selection and mutation is:
(a) symbolic reasoning (b) Bayesian inference
(c) backpropagation (d) evolutionary search

4. Given a goal, a system seeks facts that prove it. That is:
(a) backward chaining (b) forward chaining
(c) model-based control (d) gradient descent

5. The first AI winter began about 1974 because of:
(a) the GPU (b) a report, fund cuts
(c) the Dartmouth meeting (d) Turing's death

6. Which part of an expert system explains a conclusion?
(a) knowledge base (b) acquisition module
(c) explanation facility (d) inference engine

7. Deep Blue beating Kasparov in 1997 is:
(a) weak AI (b) strong AI
(c) consciousness (d) first-order logic

8. AlphaGo (2016) combined deep learning with:
(a) if–then rules (b) a knowledge base
(c) Monte Carlo tree search (d) semantic networks

9. STRIPS is an example of:
(a) knowledge representation (b) automated planning
(c) computer vision (d) evolutionary search

10. A single-layer perceptron cannot learn:
(a) AND (b) NOT
(c) XOR (d) OR

Answer key → 1-b · 2-c · 3-d · 4-a · 5-b · 6-c · 7-a · 8-c · 9-b · 10-c

Written question: Define AI. Explain Turing's test, name the four families of AI approach with one example each, and account for the two AI winters. 10 marks

1. **Definition (2):** AI builds systems that perform tasks normally needing human intelligence — reasoning, learning, perception, planning. 2
2. **Turing test (2):** Turing, 1950. An interrogator holds a written conversation with a hidden human and a hidden machine; the machine passes if he cannot reliably tell which is which. Conditions: natural language, no prior information, a third human as safety valve. 2
3. **Four families (3):** symbolic — expert systems such as MYCIN, knowledge in written rules; connectionist — neural networks, knowledge in weights fitted to data; evolutionary — genetic algorithms, knowledge in the fittest of a population; probabilistic — Bayesian networks and HMM. 3
4. **The winters (3):** 1974–1980, after the Lighthill report and funding cuts; 1987–1993, when the expert-system boom collapsed with the Lisp machine market. 3

Before the exam ✓

- ✓ Can give 1950 / 1955 / 1956
- ✓ Can state the Turing test
- ✓ Can name four families + examples
- ✓ Can write an agent problem with PEAS